Linguistic Universals as Quantale-Enriched Cognition–Language Correspondences: Causal Coding, TUG, and Weakest Explanations

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Abstract

The paper previously formulated a structural correspondence between two bodies of work: a TyLAA/TUG account of real-world linguistic universals and a causal-coding account of continual learning. The present version strengthens that correspondence by giving it a category-theoretic and quantale-weakness formulation. The central claim is this:

$$\boxed{\begin{array}{c}\text{linguistic universals are direct images of cognitive invariants}\\ \text{along a weakest, quantale-enriched externalization correspondence}\end{array}}$$

and conversely

$$\boxed{\begin{array}{c}\text{cognitive universals are weakest pullback explanations}\\ \text{of stable linguistic subobjects.}\end{array}}$$

For the non-specialist, the first sentence says that robust cross-linguistic patterns are not merely correlations among surface features: they are the observable shadows of stable cognitive patterns, transported through the machinery that turns cognition into language. The second sentence says that, when we start from the typological record, the best explanation of a linguistic universal is the weakest cognitive structure that has to be posited in order to make that linguistic pattern stable.

We make this precise by introducing three quantale-enriched categories: a category Cog_Q of causally factorized cognitive learners, a category Frame_Q of semantic frames, and a category Lang_Q of TUG trace quotients under typological interfaces. The cognition-to-language map is represented by a lax symmetric monoidal enriched functor

$$\Lambda = \mathcal{E} \circ \mathcal{K} : \mathsf{Cog}_Q \longrightarrow \mathsf{Lang}_Q,$$

where $\mathcal{K}$ extracts semantic-frame structure from a cognitive substrate and $\mathcal{E}$ externalizes semantic frames into linguistic cores. In the more general, non-deterministic case the same relation is a Q-enriched profunctor $\mathcal{P} : \mathsf{Cog}_Q^{op} \times \mathsf{Lang}_Q \to Q$. Linguistic universals become subobjects or predicates in Lang_Q; cognitive explanations become pullbacks of those subobjects along Λ, selected by a quantale-valued weakness criterion that penalizes unnecessary cost, confusion, and double-counting.

This categorical formulation turns the earlier five correspondences into theorem-shaped claims. The strongest is a monoidal transport theorem: if cognitive context-updates act on disjoint causal factors, then their linguistic images are swap-equivalent in the TUG trace quotient; if they nearly commute, the linguistic swap defect is bounded by the image of the cognitive commutator defect. Hence the TUG factorization theorem and the causal-continual-learning commutator theorem are two instances of one enriched monoidal fact. The other four

correspondences become closure transport, sparse-energy pushforward, stability-rate transport, and context-indexed gluing theorems. Quantale weakness then supplies the selection principle: causal explanations beat redundant correlation-only explanations because they introduce the weakest mediator and avoid double-counting the same dependency through multiple paths.

Contents

1 Introduction

1.1 The new main message

The original version of this paper argued that the five main conclusions of the TUG-based paper on linguistic universals [1] line up closely with the five main architectural features of causal-coding continual learning [4, 5]. This version keeps that thesis but changes the mathematical frame. The correspondence is no longer treated as a list of analogies. It is formulated as a structured map between categories:

$$\text{cognitive causal systems} \quad \longrightarrow \quad \text{semantic frames} \quad \longrightarrow \quad \text{linguistic TUG cores.}$$

The result is a more compact and more powerful statement:

$$\boxed{\begin{array}{c}\text{linguistic universals are direct images of cognitive invariants}\\ \text{along a weakest, quantale-enriched externalization correspondence}\end{array}}$$

and conversely

$$\boxed{\begin{array}{c}\text{cognitive universals are weakest pullback explanations}\\ \text{of stable linguistic subobjects.}\end{array}}$$

These are deliberately strong sentences, but they should not remain Delphic. Let us unpack them before giving the math.

A *cognitive invariant* is a pattern in the cognitive substrate that remains stable across contexts. Examples, at the level of abstraction relevant here, include person-role frames, number frames, argument-role frames, causal-event frames, and the modular decomposition that prevents one context of learning from destroying another. A *linguistic universal* is an observable regularity in the TUG core of human languages: for example, a person hierarchy, a number hierarchy, a case hierarchy, a word-order coherence tendency, or a context-local grammatical law.

A *direct image* is the mathematical version of pushing a pattern forward through a structure-preserving map. If a cognitive pattern is stable inside the mind, and if the language-production system preserves the relevant structure, then the image of that cognitive pattern becomes a linguistic regularity. A *pullback* is the mathematical version of taking a linguistic pattern and asking what structure must have existed upstream in order for this pattern to be observed downstream. A *weakest pullback explanation* is the least overcommitted such upstream explanation: the one that explains the linguistic pattern while positing the fewest unnecessary distinctions, couplings, hidden variables, or double-counted correlations.

The word *quantale* supplies the quantitative Occam principle. A quantale is an algebraic structure that lets us combine and compare values such as weakness, cost, confidence, confusion, and double-counting. It lets us say, not just informally but algebraically, that one explanation is weaker than another because it preserves more indistinction while using less cost and fewer redundant dependencies. This is the version of weakness developed in [8]; it extends the TyLAA/TUG idea of choosing the weakest red operational system consistent with a blue type interface [7, 2] and composes it with causal-coding weakness and with the independence-versus-dependency Pareto frontier of [6].

1.2 Why category theory is the right language here

The core problem is not merely to compare two sets of features. Linguistic universals live in a space of derivations, interfaces, quotients, symmetries, context restrictions, and diachronic transitions. Cognitive universals live in a space of causal modules, update endomorphisms, semantic frames, context supports, and stability gradients. The map between them is not a function from one list to another; it must preserve composition, modularity, locality, and weakness. Category theory is designed for exactly this kind of situation.

Conceptually:

- A **category** is a world of objects and structure-preserving arrows between them.
- A **functor** is a map between such worlds that preserves arrows and their composition.
- A **monoidal category** is a category in which objects can be composed side by side; here, tensor product represents modular factorization.
- A **lax monoidal functor** preserves this side-by-side composition up to controlled slack. This matters because cognition-to-language externalization preserves modularity only approximately.
- An **enriched category** replaces yes/no arrows by arrows valued in a structure such as a quantale. This lets morphisms carry weakness, cost, confidence, and error.
- A **pullback** is the categorical preimage of a pattern. It tells us what upstream structure is required for a downstream pattern.
- A **direct image** pushes an upstream pattern forward into the downstream world.
- An **adjunction** formalizes a pair of translations that are two sides of one process; here, production and comprehension are modeled as adjoint operations between semantic frames and linguistic forms.

These notions are not decorative. They let us state the central thesis as a precise transport theorem rather than as a suggestive comparison.

1.3 The three-layer correspondence

The basic diagram of the paper is:

$$\mathsf{Cog}_Q \xrightarrow{\mathcal{K}} \mathsf{Frame}_Q \xrightarrow{\mathcal{E}} \mathsf{Lang}_Q.$$

Here Cog_Q is the category of causal-coding cognitive systems; Frame_Q is the category of semantic frames; and Lang_Q is the category of TUG linguistic cores. The functor $\mathcal{K}$ extracts from a cognitive system the frames it supports, while $\mathcal{E}$ externalizes those frames into linguistic structure. Their composite

$$\Lambda := \mathcal{E} \circ \mathcal{K} : \mathsf{Cog}_Q \to \mathsf{Lang}_Q$$

is the cognition-to-language correspondence.

The semantic-frame layer is important. It is too rigid to map directly from neural or cognitive modules to surface linguistic features. Between them lie semantic frames: event frames, force-dynamic frames, argument-role frames, person/number frames, possession frames, attention frames, and other frame-like cognitive schemas. The general theory of general intelligence in [9] treats mind as a network of patterns and emphasizes cognitive synergy among different kinds of cognitive process. The frame category used here is a formal bridge between that patternist/cognitive-synergy picture and the TUG machinery of grammar space.

1.4 The empirical motivation

The empirical motivation remains the same as in the original draft. The formal TUG paper [1] joins a TyLAA-based account of grammar cores with large-scale statistical evidence about proposed implicational universals. The empirical pattern is striking: hierarchical universals survive correction best; narrow word-order universals survive moderately; broad cross-module universals survive poorly unless mediated; universalhood is graded rather than binary; and many genuine universals are context-indexed local laws rather than sweeping global laws.

The causal-coding papers derive, from continual-learning considerations, the same kind of architecture: protected stable substrate, modular factorization, sparse coupling, graded kernel-shell stability, and context-conditioned routing [4, 5]. The claim of this paper is that the match is not accidental. If language is carried by a cognitive architecture that has to learn many contexts without catastrophic forgetting, then its typological footprint should look exactly like the corrected linguistic-universals record.

1.5 What is new in this version

The paper now contributes the following category-theoretic refinements.

1. It defines Q-enriched categories Cog_Q, Frame_Q, and Lang_Q, together with a lax monoidal correspondence $\Lambda = \mathcal{E} \circ \mathcal{K}$.
2. It treats linguistic universals as subobjects in Lang_Q, cognitive invariants as subobjects in Cog_Q, and the cognition-language relation as direct image and pullback between subobject lattices.
3. It defines a category $\mathrm{Exp}(U)$ of explanations of a linguistic universal U, and selects the weakest explanation using quantale weakness.
4. It proves a monoidal transport theorem showing that causal modularity maps to TUG modularity and that commutator defects map to swap defects.
5. It formulates mediators as residuals, i.e. weakest contexts making a cross-module implication true.
6. It reformulates the five earlier correspondences as closure transport, factorization, sparse-energy pushforward, stability-rate transport, and context gluing.

2 Background: TUG, causal coding, and quantale weakness

2.1 TUG as the weakest typological grammar core

The TUG formalism begins with an operational grammar G_ℓ for a language ℓ. Its derivations form a space $\mathrm{Deriv}(G_\ell)$. One then specifies an observation interface, an independence relation, and a symmetry group. For typology the observation interface must record at least logical form, phonological/externalization information, and morphosyntactic information:

$$\mathcal{O}_{\mathrm{typ}} = (\Sigma_{\mathrm{LF}}, \Sigma_{\mathrm{PF}}, \Sigma_{\mathrm{Morph}}).$$

The typological TUG core is the quotient

$$C_\ell := \mathrm{TUG}_{\mathcal{O}_{\mathrm{typ}}}(G_\ell) := \mathrm{Deriv}(G_\ell)/{\equiv_{\perp,\Gamma,\mathcal{O}_{\mathrm{typ}}}},$$

where $\equiv_{\perp,\Gamma,\mathcal{O}_{\mathrm{typ}}}$ is the maximal sound congruence generated by swapping adjacent independent actions, applying symmetries, and identifying derivations that the interface cannot distinguish.

For non-specialists, the quotient says: if two derivations are indistinguishable by the typological interface, then the grammar core should not keep them apart. The core is the weakest operational system that still preserves what typology can observe. This is the natural-language analogue of the TyLAA principle that, given a blue type/proof interface, one chooses the weakest red operational explanation consistent with it [7, 2].

2.2 The five TUG universals conclusions

The TUG-based linguistic-universals paper [1] extracts five mathematical conclusions from the empirical pattern of which proposed universals survive correction.

C1. Hierarchical universals are closure systems. A hierarchy of implications generates a closure operator cl_R on a feature set. Admissible paradigms are fixed points of that operator, equivalently order ideals after quotienting mutual implication.

C2. Cross-module universals require mediation. If two grammatical modules are independent in the TUG sense, the trace quotient factors; no nontrivial implication can cross from one factor to another unless a mediator breaks the factorization.

C3. Narrow word-order universals are low-frustration externalization laws. Word-order regularities are modeled not by one global head-direction bit but by a sparse signed graph of local externalization choices.

C4. Universalhood is graded by dynamical stability. A proposed universal should be assigned a stability coefficient derived from transition rates rather than a binary supported/unsupported label.

C5. Many genuine universals are context-indexed local laws. Broad unconditional claims often fail because they pool over distinct grammatical contexts in which real local laws hold.

These conclusions already have a categorical flavor. Closure systems are algebraic subobject structures; factorization is a monoidal statement; low-frustration word order is an energy pushforward problem; stability is a functorial comparison of dynamics; context-indexed laws are sheaf-like gluing problems. The rest of the paper makes this explicit.

2.3 Causal coding and the General Causal-Continual-Learning theorem

Causal coding begins from a continual-learning problem. A learner encounters a stream of contexts $c \in C$. Each context induces an update operator U_c on the learner's internal state. If all contexts rewrite the same entangled parameters, catastrophic forgetting is unavoidable. If the internal state factors into causal modules and each context updates mostly the modules it actually needs, then the updates approximately commute, and sequential learning approximates joint learning.

In its categorical form, the Causal-Continual-Learning theorem considers a monoidal object

$$A \cong A_1 \otimes \cdots \otimes A_n$$

and update endomorphisms

$$U_c : A \to A$$

with supports $S_c \subseteq \{1, \ldots, n\}$. If U_c acts as the identity outside S_c, then updates with disjoint supports commute up to the canonical monoidal symmetry:

$$S_c \cap S_d = \emptyset \quad \Longrightarrow \quad U_c \circ U_d \cong U_d \circ U_c.$$

The approximate version measures deviations by a metric or a quantale-valued defect; the total path-dependence is bounded by a sum over genuinely confusing context pairs [4]. This is the cognitive-learning counterpart of the TUG trace-factorization theorem.

2.4 HBCML and kernel-shell stability

HBCML and ColBaC make the CCL theorem constructive [5]. They propose a two-level architecture:

- a sparse top-level controller selecting a subset of modules or columns for each context;
- internal probabilistic structure within each column, distinguishing protected hard-kernel material, reusable inner-shell structure, semi-general middle-shell structure, and task-local outer-shell residue.

The hard kernel is non-prunable; inner shells are difficult to perturb; middle shells are adaptable; outer shells are disposable. This is a natural architectural correlate of the graded stability coefficient proposed for universals in [1].

2.5 Quantale weakness

The weakness framework [8] generalizes Occam's razor from syntactic shortness to semantic under-commitment. In its simplest form, a hypothesis is weak if it identifies many pairs that do not need to be distinguished. In the quantale formulation, one chooses a commutative quantale $(Q, \leq, \otimes, e, \bigvee)$, a set X, and a valuation $\mu : X \to Q$. For a relation $H \subseteq X \times X$ interpreted as indistinction, define

$$w(H) := \bigvee_{(x,y) \in H} \mu(x) \otimes \mu(y).$$

Larger $w(H)$ means that the relation treats more important pairs as indistinct and therefore makes fewer unnecessary distinctions. This formula recovers Bennett-style set weakness, probabilistic weakness, logical-entropy/graphtropy-style weakness, and other variants by changing the quantale and valuation.

For the present paper we need a product quantale that tracks several desiderata at once.

Definition 2.1 (Explanatory weakness quantale)**.** Let

$$Q = Q_{\text{ind}} \times Q_{\text{fit}} \times Q_{\text{cost}}^{op} \times Q_{\text{conf}}^{op} \times Q_{\text{dc}}^{op},$$

where Q_{ind} measures indistinction/generalization, Q_{fit} measures empirical fit or confidence, Q_{cost} measures computational or description cost, Q_{conf} measures commutator/confusion cost, and Q_{dc} measures double-counting. The superscript *op* reverses the order in the cost-like components, so that lower cost, lower confusion, and lower double-counting are better. The product is ordered componentwise and has componentwise monoidal product and joins.

The interpretation is simple: an explanation is better, hence weaker in the generalized Occam sense, if it preserves more admissible indistinction, fits the data at least as well, uses less cost, causes less context confusion, and double-counts fewer dependencies.

2.6 The independence/dependency Pareto frontier

The tradeoff paper [6] argues that practical intelligence lives on a Pareto frontier between two routes:

Route A: transform the representation so that channels become more independent;

Route B: explicitly model dependencies among channels.

Causal explanations are favored in this framework because they often identify a small number of genuine mediators instead of simulating the same dependence through many overlapping correlations. In the quantale above, the latter incurs a larger DC component and often a larger Conf component. Thus the weakness principle does not merely say "prefer simpler explanations" in prose; it supplies an algebraic order in which causal explanations can dominate redundant correlation-only explanations.

3 The three enriched categories

We now define the categorical setting. The definitions are intentionally somewhat abstract. They are meant to isolate the structure needed for the theorems rather than to encode every implementation detail of every possible cognitive or linguistic system.

3.1 A note on enriched and weighted categories

For technical convenience, we use a lightweight version of enrichment. A Q-weighted category is an ordinary category whose arrows carry grades in Q, with composition respecting those grades. When Q is viewed as a one-object monoidal category, this is a familiar form of enrichment; when the arrows are ordinary maps plus explicit defect bounds, it is often more readable to call the result a weighted category.

Definition 3.1 (Q-weighted category)**.** A Q-weighted category $\mathcal{C}$ consists of an ordinary category, together with a grade function

$$\omega_{X,Y} : \mathcal{C}(X, Y) \to Q$$

such that

$$e \leq \omega(\mathrm{id}_X), \qquad \omega(g) \otimes \omega(f) \leq \omega(g \circ f)$$

whenever $f : X \to Y$ and $g : Y \to Z$. We read $\omega(f)$ as the weakness/quality grade of the morphism f. The product quantale order is chosen so that larger grades are better.

For readers unfamiliar with this language: the grade says how good a structure-preserving map is. In a strict category an arrow either exists or not. Here an arrow can exist with a quantitative quality profile: how much structure it preserves, how much error it introduces, how much cost it incurs, and so on.

3.2 The cognitive category Cog_Q

Definition 3.2 (Causal-coding cognitive object)**.** An object $A \in \mathsf{Cog}_Q$ is a tuple

$$A = (\Theta_A \cong \Theta_{A,1} \otimes \cdots \otimes \Theta_{A,n_A}, C_A, \{U_c : A \to A\}_{c \in C_A}, \{S_c \subseteq \{1, \ldots, n_A\}\}_{c \in C_A}, \mu_A),$$

where:

- Θ_A is the internal state or parameter object;
- $\Theta_{A,i}$ are causal modules;
- C_A is a set or category of contexts;
- U_c is the update endomorphism induced by context c;
- S_c is the support of context c, i.e. the set of modules it is allowed to touch nontrivially;
- μ_A is a valuation assigning weakness/cost/importance grades to states, modules, frames, or update traces.

A morphism in Cog_Q is a causal simulation: it maps one causal learner into another while preserving context-conditioned updates up to controlled defect.

Definition 3.3 (Causal simulation morphism)**.** A morphism

$$f : A \to B$$

in Cog_Q consists of a map of state objects $f_\Theta : \Theta_A \to \Theta_B$ and a map of contexts $f_C : C_A \to C_B$, such that for each $c \in C_A$ the square

$$\begin{array}{ccc} \Theta_A & \xrightarrow{U_c} & \Theta_A \\ {\scriptstyle f_\Theta}\big\downarrow & & \big\downarrow{\scriptstyle f_\Theta} \\ \Theta_B & \xrightarrow[U_{f_C(c)}]{} & \Theta_B \end{array}$$

commutes up to a quantale-valued defect

$$\delta_c(f) := d_B\big(f_\Theta \circ U_c, U_{f_C(c)} \circ f_\Theta\big).$$

The grade $\omega(f) \in Q$ is obtained by aggregating the indistinction preserved by f, empirical fit, cost, confusion defects $\delta_c(f)$, and double-counting costs.

Conceptually, a morphism $A \to B$ says: the second cognitive system can simulate the first while preserving its causal update structure. The square says that it should not matter whether one first updates in context c and then maps forward, or first maps forward and then updates in the corresponding context. The defect measures the failure of this equality.

3.3 The semantic-frame category Frame_Q

Definition 3.4 (Semantic-frame category)**.** The category Frame_Q has as objects semantic frames, including event frames, argument-role frames, force-dynamic frames, person and number frames, possession frames, spatial frames, attention frames, and other cognitive schemas relevant to linguistic expression. A morphism

$$r : F \to F'$$

is a frame refinement, abstraction, embedding, or transformation, graded by how much semantic structure it preserves, how much it abstracts away, and how costly it is to represent.

A semantic frame is the bridge between cognition and language. A person hierarchy is not merely a syntactic pattern; it is an externalized trace of a frame distinguishing speech-act participants and third parties. A case hierarchy is not merely a list of morphological markings; it is an externalized trace of an argument-role frame. The frame category is where these cognitive patterns are represented before they become linguistic observations.

3.4 The linguistic category Lang_Q

Definition 3.5 (TUG linguistic object)**.** An object $L \in \mathsf{Lang}_Q$ is a typological TUG core

$$L = (M_L, \mathrm{Lab}_L, \perp_L, \Gamma_L, \mathcal{O}_{\mathrm{typ},L}, \mu_L)\,,$$

where

$$M_L = \mathrm{Lab}_L^*/\equiv_{\perp,\Gamma,\mathcal{O}_{\mathrm{typ}}}$$

is the trace quotient of visible grammatical actions by the maximal sound typological congruence. The interface

$$\mathcal{O}_{\mathrm{typ},L} = (\Sigma_{\mathrm{LF}}, \Sigma_{\mathrm{PF}}, \Sigma_{\mathrm{Morph}})$$

records the information needed for typological features. The valuation μ_L grades derivations, features, universals, or trace pairs.

Definition 3.6 (Linguistic morphism)**.** A morphism

$$h : L \to L'$$

in Lang_Q is an interface-preserving trace functor. It maps labels, traces, symmetries, independence relations, and typological predicates from L to L', preserving observations up to defect:

$$\begin{tikzcd}
M_L \arrow[r, "h"] \arrow[d, "\mathcal{O}_L"'] & M_{L'} \arrow[d, "\mathcal{O}_{L'}"] \\
\mathrm{Obs}_L \arrow[r, "\mathrm{Obs}(h)"'] & \mathrm{Obs}_{L'}\,.
\end{tikzcd}$$

The grade $\omega(h) \in Q$ records how much typological information, independence structure, symmetry structure, and weakness are preserved.

3.5 Pattern categories and subobjects

Universals are not usually whole languages or whole cognitive systems. They are patterns inside them. Categorically, patterns are represented by subobjects.

Definition 3.7 (Pattern as subobject)**.** For an object X in one of the categories above, a pattern in X is a subobject

$$P \hookrightarrow X.$$

The poset of subobjects of X is denoted $\mathrm{Sub}(X)$. A linguistic universal in a language core L is a subobject of $\mathrm{Obs}(L)$, or equivalently a predicate on L's typological core. A cognitive invariant in A is a subobject $I \hookrightarrow A$ stable under the relevant update endomorphisms.

For less mathematical readers: a subobject is the categorical version of a subset, property, or predicate. The universal "if a language has a dual, it has a plural" is not a language; it is a constraint on possible feature states. That constraint is represented as a subobject of the feature-state space.

4 The cognition–language correspondence

4.1 The representable case: a functor

In the most direct case, the cognition-language correspondence is a functor

$$\Lambda : \mathsf{Cog}_Q \to \mathsf{Lang}_Q$$

factoring through frames:

$$\begin{tikzcd} \mathsf{Cog}_Q \arrow[r, "\mathcal{K}"] \arrow[rr, bend right, "\Lambda"'] & \mathsf{Frame}_Q \arrow[r, "\mathcal{E}"] & \mathsf{Lang}_Q. \end{tikzcd}$$

Here $\mathcal{K}$ extracts from a cognitive system the semantic frames supported by its causal modules, and $\mathcal{E}$ externalizes frames into TUG trace structure. The composite $\Lambda = \mathcal{E} \circ \mathcal{K}$ maps a cognitive substrate to its linguistic externalization core.

Definition 4.1 (Cognition–language functor)**.** A cognition–language functor is a Q-weighted lax symmetric monoidal functor

$$\Lambda : \mathsf{Cog}_Q \to \mathsf{Lang}_Q$$

factoring as $\mathcal{E} \circ \mathcal{K}$, together with structural morphisms

$$\lambda_{A,B} : \Lambda(A) \otimes \Lambda(B) o \Lambda(A \otimes B), \qquad \lambda_0 : I_{\mathsf{Lang}_Q} \to \Lambda(I_{\mathsf{Cog}_Q}),$$

whose grades record the degree to which linguistic externalization preserves cognitive factorization.

The adjective *lax* matters. Cognitive modules do not always externalize into perfectly independent linguistic modules. Sometimes externalization introduces couplings; sometimes it forgets distinctions; sometimes it pushes several cognitive factors through the same phonological or morphological channel. Lax monoidality expresses "preserves modular composition, but possibly with slack."

4.2 The non-representable case: a profunctor

A single cognitive system may support multiple linguistic externalizations, and a single linguistic pattern may be supported by multiple cognitive substrates. In that case a functor may be too deterministic. The more general object is a profunctor.

Definition 4.2 (Cognition–language profunctor)**.** A general cognition–language correspondence is a Q-enriched profunctor

$$\mathcal{P} : \mathsf{Cog}_Q^{op} \times \mathsf{Lang}_Q \to Q.$$

The value $\mathcal{P}(A, L) \in Q$ grades how well cognitive system A explains or realizes linguistic core L. The functorial case is recovered when

$$\mathcal{P}(A, L) = \mathsf{Lang}_Q(\Lambda A, L).$$

Conceptually, the profunctor says: instead of assigning one language core to one cognitive system, we assign a graded space of possible correspondences. This is often more realistic. Human cognition, typological data, and semantic frames are underdetermined in both directions.

4.3 Production and comprehension as an adjunction

The factorization $\Lambda = \mathcal{E} \circ \mathcal{K}$ separates frame extraction from linguistic externalization. The externalization functor $\mathcal{E} : \mathsf{Frame}_Q \to \mathsf{Lang}_Q$ should have an interpretation or comprehension partner

$$\mathcal{S} : \mathsf{Lang}_Q \to \mathsf{Frame}_Q.$$

The natural formal relation is an adjunction:

$$\mathcal{E} \dashv \mathcal{S}.$$

Definition 4.3 (Externalization–interpretation adjunction)**.** An externalization–interpretation adjunction consists of functors

$$\mathcal{E} : \mathsf{Frame}_Q \to \mathsf{Lang}_Q, \qquad \mathcal{S} : \mathsf{Lang}_Q \to \mathsf{Frame}_Q$$

and a natural isomorphism of hom-objects

$$\mathsf{Lang}_Q(\mathcal{E}F, L) \cong \mathsf{Frame}_Q(F, \mathcal{S}L).$$

The unit and counit are

$$\eta_F : F \to \mathcal{S}\mathcal{E}F, \qquad \epsilon_L : \mathcal{E}\mathcal{S}L \to L.$$

The adjunction means: producing a linguistic object from a semantic frame is equivalent, in the appropriate categorical sense, to interpreting that linguistic object as a semantic frame. The unit and counit express the two round-trips:

$$F \xrightarrow{\eta_F} \mathcal{S}\mathcal{E}F \qquad \mathcal{E}\mathcal{S}L \xrightarrow{\epsilon_L} L.$$

For readers with a linguistics background: this is a formal version of the production/comprehension duality. It does not say that production and comprehension are identical processes. It says their correctness conditions are paired by a universal relation.

4.4 Direct images and pullbacks of universals

Now we can state the core thesis precisely. Assume that the subobject fibrations over Cog_Q and Lang_Q admit pullback along Λ, and, on the relevant sublattices, direct image Σ_Λ left adjoint to pullback:

$$\Sigma_\Lambda \dashv \Lambda^*.$$

Thus for subobjects $I \hookrightarrow A$ and $U \hookrightarrow \Lambda A$,

$$\Sigma_\Lambda(I) \le U \iff I \le \Lambda^*(U).$$

Definition 4.4 (Direct image of a cognitive invariant)**.** Let $I \hookrightarrow A$ be a cognitive invariant. Its linguistic direct image is

$$\Sigma_\Lambda(I) \hookrightarrow \Lambda A.$$

When $\Sigma_\Lambda(I)$ is stable under the relevant linguistic dynamics, it is a linguistic universal induced by the cognitive invariant I.

Definition 4.5 (Pullback explanation of a linguistic universal)**.** Let $U \hookrightarrow L$ be a linguistic universal and let $\eta : \Lambda A \to L$ be a realization map. The cognitive pullback explanation of U along $\eta \circ \Lambda$ is the subobject $P_A \hookrightarrow A$ given by the pullback square

$$\begin{tikzcd} P_A \arrow[r] \arrow[d, hook] & U \arrow[d, hook] \\ A \arrow[r, "\eta\circ\Lambda"'] & L. \end{tikzcd}$$

A weakest pullback explanation is a maximal-weakness subobject among the pullback explanations compatible with the evidence.

This diagram is the precise form of the second boxed sentence from the abstract. A linguistic universal is a subobject U of the linguistic world. Pulling it back gives the cognitive structure required to produce it. Weakness then chooses the least overcommitted such structure.

4.5 Invariants transport to universals

Definition 4.6 (Invariant subobject)**.** Let $A \in \mathsf{Cog}_Q$. A subobject $I \hookrightarrow A$ is invariant under a family of updates $\{U_c\}_{c\in C}$ if for every c there exists an endomorphism $U_c|_I : I \to I$ making the square commute:

$$\begin{tikzcd} I \arrow[r, "U_c|_I"] \arrow[d, hook] & I \arrow[d, hook] \\ A \arrow[r, "U_c"'] & A. \end{tikzcd}$$

In the approximate/enriched case the square commutes up to a defect bounded in Q.

Theorem 4.7 (Direct image of invariants)**.** *Let $\Lambda : \mathsf{Cog}_Q \to \mathsf{Lang}_Q$ be a Q-weighted functor that maps cognitive updates U_c to linguistic actions $\Lambda(U_c)$ non-expansively with respect to invariance defects. If $I \hookrightarrow A$ is invariant under $\{U_c\}$, then $\Sigma_\Lambda(I) \hookrightarrow \Lambda A$, when it exists, is invariant under $\{\Lambda(U_c)\}$. In the approximate case, the linguistic invariance defect is bounded by the image under Λ's defect map of the cognitive invariance defect.*

Proof. In the strict case, invariance of I gives a commuting square for each U_c. Functoriality maps that square to a commuting square in Lang_Q. Direct image along Λ preserves the corresponding subobject inclusion by the adjunction $\Sigma_\Lambda \dashv \Lambda^*$. Thus $\Sigma_\Lambda(I)$ is stable under $\Lambda(U_c)$. In the enriched case, each square has a defect grade; non-expansiveness of Λ maps the defect to a no-larger linguistic defect, giving the stated bound. □

This theorem is the precise form of the claim that linguistic universals are direct images of cognitive invariants.

5 Weakest explanations and quantale selection

5.1 The explanation category

A linguistic universal generally has many possible explanations. Some are causal and sparse; others are correlation-based and redundant. We represent these possibilities categorically.

Definition 5.1 (Explanation category)**.** Let $U \hookrightarrow L$ be a linguistic universal. The category $\mathrm{Exp}(U)$ has as objects tuples

$$E = (A, F, L_E, \eta, \rho),$$

where

- $A \in \mathsf{Cog}_Q$ is a cognitive causal substrate;
- $F = \mathcal{K}(A) \in \mathsf{Frame}_Q$ is its extracted semantic-frame structure;
- $L_E = \mathcal{E}(F) \in \mathsf{Lang}_Q$ is its linguistic externalization;
- $\eta : L_E \to L$ is an interface-respecting realization map;
- $\rho : U \hookrightarrow L$ is witnessed by the image of L_E under η.

A morphism $E \to E'$ is a structure-preserving transformation of explanations commuting with the realization maps up to Q-defect.

For each explanation E choose a universe X_E of relevant states, frames, traces, or feature configurations. Let

$$H_E \subseteq X_E \times X_E$$

be the indistinction relation induced by E: $(x, y) \in H_E$ means that explanation E does not distinguish x and y. Its base weakness is

$$w(H_E) = \bigvee_{(x,y)\in H_E} \mu_E(x) \otimes \mu_E(y).$$

Definition 5.2 (Explanation weakness)**.** The full weakness grade of an explanation E is the product-quantale value

$$W(E) = \big(w(H_E), \mathrm{Fit}(E), \mathrm{Cost}(E), \mathrm{Conf}(E), \mathrm{DC}(E)\big) \in Q_{\mathrm{ind}} \times Q_{\mathrm{fit}} \times Q^{op}_{\mathrm{cost}} \times Q^{op}_{\mathrm{conf}} \times Q^{op}_{\mathrm{dc}}.$$

A weakest explanation of U is any

$$E^* \in \arg\max_{E\in\mathrm{Exp}(U)} W(E).$$

Because the cost, confusion, and double-counting components are ordered oppositely, maximizing W means maximizing indistinction and fit while minimizing cost, confusion, and double-counting.

5.2 Relation to TyLAA weakness

TyLAA selects the weakest red operational explanation consistent with a blue type/proof interface. Here we select the weakest cognitive causal explanation consistent with a linguistic universal. These are not two unrelated Occam principles. They are two layers of the same quantale-enriched selection process.

Definition 5.3 (Weakness-preserving correspondence)**.** Let W_C be a cognitive weakness measure and W_L a linguistic/TUG weakness measure. A cognition-language functor Λ is weakness-preserving if there exists a quantale homomorphism

$$\varphi : Q_C \to Q_L$$

such that for every cognitive indistinction relation H,

$$W_L(\Lambda H) \geq \varphi(W_C(H)),$$

with equality in the strict case.

Theorem 5.4 (Composition of TyLAA and cognitive weakness)**.** *Assume Λ is weakness-preserving. Let H_C^* be a maximally weak sound cognitive congruence compatible with the semantic-frame interface. Then $\Lambda(H_C^*)$ is maximally weak among linguistic congruences in the image of Λ. If Λ is essentially surjective on sound linguistic congruences, then the TyLAA weakest-red congruence and the image of the weakest cognitive congruence coincide.*

Proof. Let K be any sound linguistic congruence in the image of Λ, say $K = \Lambda(H)$. Since H_C^* is maximally weak among compatible cognitive congruences, $W_C(H) \leq W_C(H_C^*)$. Applying the quantale homomorphism and weakness preservation gives

$$W_L(K) = W_L(\Lambda H) \leq W_L(\Lambda H_C^*)$$

within the image. If Λ is essentially surjective on sound linguistic congruences, every sound linguistic congruence is isomorphic to one in the image, so the same maximality holds globally. □

This theorem is the formal version of the intuition that TyLAA weakness should compose with causal-coding weakness rather than sit beside it as a separate principle.

5.3 Why causal explanations beat double-counting explanations

The quantale order makes precise why causal explanations are preferred over redundant correlation stories.

Proposition 5.5 (Causal explanations dominate redundant explanations)**.** *Let E_{causal} and E_{corr} be two explanations of the same linguistic universal U. Suppose they have equal or comparable fit and that*

$$w(H_{E_{\text{causal}}}) \geq w(H_{E_{\text{corr}}}),$$

$$\mathrm{Cost}(E_{\text{causal}}) \leq \mathrm{Cost}(E_{\text{corr}}), \qquad \mathrm{Conf}(E_{\text{causal}}) \leq \mathrm{Conf}(E_{\text{corr}}), \qquad \mathrm{DC}(E_{\text{causal}}) < \mathrm{DC}(E_{\text{corr}}).$$

Then

$$W(E_{\text{causal}}) > W(E_{\text{corr}})$$

in the product quantale order, provided at least one of the displayed inequalities is strict in a non-ignored component.

Proof. This is immediate from the componentwise product order and the use of opposite order in the cost-like components. Equal or better fit, greater or equal indistinction, lower or equal cost/confusion, and strictly lower double-counting imply strict dominance. □

The proposition is simple, but it is important. A causal explanation wins not because "causal" is a magic word, but because a genuine mediator lets the theory represent one dependency once. A correlation-only explanation often has to represent the same dependency through many overlapping pairwise links, increasing double-counting and confusion.

6 Main theorem: monoidal transport of causal modularity

6.1 Commutators and swap defects

Let $A \in \mathsf{Cog}_Q$ have context updates $U_c, U_d : A \to A$. Define their cognitive commutator defect by

$$\mathrm{Comm}_C(c,d) := d_C(U_c \circ U_d, U_d \circ U_c).$$

Let $L = \Lambda A$. The images $\Lambda(U_c), \Lambda(U_d)$ are linguistic actions or trace transformations. Their linguistic swap defect is

$$\mathrm{Comm}_L(c,d) := d_L(\Lambda(U_c) \circ \Lambda(U_d), \Lambda(U_d) \circ \Lambda(U_c)).$$

The functor Λ is non-expansive with respect to commutator defects if there is a quantale homomorphism $\varphi : Q_C \to Q_L$ such that

$$\mathrm{Comm}_L(c,d) \le \varphi(\mathrm{Comm}_C(c,d)).$$

The inequality is written in the defect order; equivalently, in the weakness order it says that the linguistic commutativity grade is at least the image of the cognitive commutativity grade.

6.2 The transport theorem

Theorem 6.1 (Enriched monoidal transport of causal modularity into TUG modularity)**.** *Let*

$$\Lambda : \mathsf{Cog}_Q \to \mathsf{Lang}_Q$$

be a Q-weighted lax symmetric monoidal functor, non-expansive with respect to commutator defects. Let

$$A \cong A_1 \otimes \cdots \otimes A_n$$

be a factored cognitive object with context-local update endomorphisms U_c supported on $S_c \subseteq \{1, \ldots, n\}$. Then:

1. *If $S_c \cap S_d = \emptyset$, then U_c and U_d commute up to the canonical monoidal symmetry, and their linguistic images are swap-equivalent in the TUG trace quotient.*
2. *If* $\mathrm{Comm}_C(c,d) \le \varepsilon$, *then*
$$\mathrm{Comm}_L(c,d) \le \varphi(\varepsilon).$$
3. *If $A = A_I \otimes A_J$ and Λ is strong monoidal on this split, then*
$$\Lambda(A) \cong \Lambda(A_I) \otimes \Lambda(A_J).$$
4. *Consequently, any nontrivial universal implication between predicates living purely on $\Lambda(A_I)$ and $\Lambda(A_J)$ requires either failure of strong monoidality on the split or a mediator object connecting the two factors.*

Proof. For (1), context-locality says that U_c acts as the identity outside S_c, and U_d acts as the identity outside S_d. If the supports are disjoint, the two endomorphisms act on different tensor factors. In a symmetric monoidal category, endomorphisms on disjoint tensor factors commute up to the canonical braiding and coherence isomorphisms. Applying Λ, functoriality sends this commutation to a commutation of linguistic trace actions. In the TUG quotient, adjacent independent actions are identified by the independence congruence, so the linguistic images are swap-equivalent.

For (2), non-expansiveness is exactly the displayed inequality.

For (3), strong monoidality on the split gives an isomorphism $\Lambda(A_I \otimes A_J) \cong \Lambda(A_I) \otimes \Lambda(A_J)$.

For (4), predicates on separate factors vary independently in a product. If an implication $\alpha(x) \Rightarrow \beta(y)$ holds for all $(x, y) \in X \times Y$, then either no x satisfies α, making the implication vacuous, or every y satisfies β, making it trivial. A nontrivial cross-factor implication therefore requires some structure beyond the product: either monoidality fails to be strong, or a mediator introduces a coupling. □

The theorem says that the TUG factorization theorem and the causal-continual-learning commutator theorem are the same theorem transported through Λ. The TUG theorem is the linguistic side of modularity; the CCL theorem is the cognitive-learning side.

6.3 The core commutative diagram

The theorem can be summarized by the following diagram, where the upper square is cognitive and the lower square is linguistic:

$$
\begin{tikzcd}
A \arrow[r, "U_c U_d"] \arrow[d, "\Lambda"'] & A \arrow[d, "\Lambda"] \\
\Lambda A \arrow[r, "\Lambda(U_c)\Lambda(U_d)"'] & \leq \varepsilon \\
A \arrow[r, "U_d U_c"] \arrow[u, "\Lambda"] & A \arrow[u, "\Lambda"']
\end{tikzcd}
$$

$$
\Lambda A \xrightarrow[\Lambda(U_d)\Lambda(U_c)]{} \Lambda A
$$

The diagram is schematic rather than a literal square in one category, but it captures the logical relation: if the two cognitive paths are close, then their two linguistic images are close; if the cognitive paths are equal because the supports are disjoint, then the linguistic paths are identified by the TUG swap congruence.

7 Mediators as residuals

7.1 Why broad universals need mediators

Broad universals try to connect features from different modules: for example, a syntactic feature to a morphological feature, or a phonological externalization choice to a semantic-role feature. If the modules factor, Theorem 6.1 says no nontrivial implication can cross the boundary. Therefore a genuine broad universal must involve a mediator.

A mediator is not just an informal third variable. Categorically, it is the weakest context that makes a cross-module implication true.

7.2 Residuals in a behavior quantale

Let M be a monoid of traces or behaviors, and let $\mathcal{P}(M)$ be its powerset quantale. Sequential composition of behaviors is

$$A \cdot B := \{ab : a \in A,\ b \in B\}.$$

The right residual is

$$A \Rightarrow C := \{m \in M : A \cdot \{m\} \subseteq C\}.$$

It satisfies the residuation law

$$A \cdot B \subseteq C \quad \Longleftrightarrow \quad B \subseteq (A \Rightarrow C).$$

Thus $A \Rightarrow C$ is the largest, hence weakest, behavior B such that $A \cdot B \subseteq C$.

Theorem 7.1 (Mediator as weakest residual context)**.** *Let P_A and P_B be behavior predicates associated with two factored modules. A mediator M makes the cross-module implication from P_A to P_B valid precisely when*

$$P_A \cdot M \subseteq P_B.$$

The weakest such mediator is the residual

$$M^* = P_A \Rightarrow P_B.$$

Proof. By the residuation law,

$$P_A \cdot M \subseteq P_B \quad \Longleftrightarrow \quad M \subseteq P_A \Rightarrow P_B.$$

Therefore $P_A \Rightarrow P_B$ is the largest behavior set satisfying the condition. Since larger behavior sets are weaker, it is the weakest mediator. □

Linguistically, this is the formal version of saying that case marking, argument-structure resolution, agreement morphology, or information-structure marking can mediate between modules. The mediator is not an arbitrary extra assumption; it is the weakest context needed to make the cross-module implication nontrivial.

7.3 A mediator diagram

The mediator breaks the product by introducing a span or cospan:

$$\begin{tikzcd} & M \arrow[dl] \arrow[dr] & \\ A & & B \end{tikzcd} \qquad \text{or} \qquad \begin{tikzcd} A \arrow[dr] & & B \arrow[dl] \\ & M & \end{tikzcd}$$

The first diagram reads M as a shared source or resource; the second reads M as a shared target or interface. Case marking is often closer to the second: syntactic and semantic-role information both feed into a morphological/externalization mediator.

8 The five correspondences as categorical transport results

We now reformulate the five correspondences from the original paper in the new categorical language. The result is not that all five are already fully proven identities. The factorization correspondence is an identity theorem. The others are conditional transport results: given explicit assumptions, cognitive structure maps to the corresponding linguistic structure.

8.1 Correspondence 1: hierarchical universals as closure transport

Conceptual version

A hierarchy such as person, number, case, or accessibility says: if a system has a more marked feature, it must have the less marked features below it. Mathematically, this is a closure system. Cognitively, the hypothesis is that the relevant hierarchy is carried by stable semantic-frame substrate: a hard-kernel or inner-shell frame whose structure is defended across contexts. Linguistically, the hierarchy is the direct image of that cognitive closure structure.

Mathematical formulation

Let P_C be a finite poset of cognitive-frame features and let

$$\mathrm{cl}_C : \mathcal{P}(P_C) \to \mathcal{P}(P_C)$$

be a closure operator generated by cognitive implications. Let P_L be the corresponding linguistic feature poset. Suppose the cognition-language correspondence induces a Galois connection

$$\Sigma_\Lambda : mathcal P(P_C) \rightleftarrows \mathcal{P}(P_L) : \Lambda^*.$$

Theorem 8.1 (Closure transport). *Assume that, on the relevant feature sublattice,*

$$\Sigma_\Lambda \Lambda^* = \mathrm{id}$$

and that $\Lambda^*\Sigma_\Lambda$ sends cl_C-closed sets to cl_C-closed sets. Define

$$\mathrm{cl}_L(S) := \Sigma_\Lambda\big(\mathrm{cl}_C(\Lambda^* S)\big).$$

Then cl_L is a closure operator on $\mathcal{P}(P_L)$. Its fixed points are precisely the linguistic feature states whose cognitive pullbacks are cl_C-closed.

Proof. Extensivity follows from the unit of the Galois connection and the extensivity of cl_C:

$$S = \Sigma_\Lambda \Lambda^* S \subseteq \Sigma_\Lambda \, \mathrm{cl}_C \, \Lambda^* S.$$

Monotonicity follows from monotonicity of Λ^*, cl_C, and Σ_Λ. For idempotence,

$$\mathrm{cl}_L(\mathrm{cl}_L(S)) = \Sigma_\Lambda \, \mathrm{cl}_C \, \Lambda^* \Sigma_\Lambda \, \mathrm{cl}_C \, \Lambda^* S.$$

By assumption, $\Lambda^*\Sigma_\Lambda \, \mathrm{cl}_C \, \Lambda^* S$ is cl_C-closed and contains $\mathrm{cl}_C \, \Lambda^* S$, hence applying cl_C does not change it on the relevant sublattice. Applying Σ_Λ gives $\mathrm{cl}_L(\mathrm{cl}_L(S)) = \mathrm{cl}_L(S)$. □

This theorem turns hierarchical linguistic universals into the direct image of cognitive closure. The hard-kernel story is a proposed source of cl_C; the TUG hierarchy is its externalized cl_L.

8.2 Correspondence 2: cross-module universals and factorization

This is Theorem 6.1 and Theorem 7.1. If the cognitive substrate factors and Λ is strong monoidal on the split, the linguistic core factors. A broad cross-module universal is then impossible except in vacuous/trivial cases. Therefore any surviving broad universal points to a mediator, and the weakest mediator is a residual.

The conceptual message is strong: broad universals are rare not because linguists failed to formulate them cleverly, but because a modular continual learner should not produce many global cross-module laws. Where broad universals survive, they are evidence for genuine bridge structure.

8.3 Correspondence 3: narrow word order as sparse-energy pushforward

Conceptual version

Word order is not one global parameter. It is a network of local externalization choices with coherence pressures among them. Cognitively, these pressures arise from sparse routing among semantic and production modules. Linguistically, they appear as a sparse signed interaction graph among word-order features.

Mathematical formulation

Let S_C be a finite or compact cognitive controller state space, and let

$$\pi : S_C \to X_L$$

map controller states to linguistic word-order configurations. Suppose the cognitive controller has a sparse signed energy

$$E_C(s) = \frac{1}{2} \sum_{i<j} a_{ij}(1 - t_{ij} s_i s_j),$$

where $a_{ij} \geq 0$ and $t_{ij} \in \{\pm 1\}$. Define the linguistic pushforward energy by Gibbs marginalization:

$$e^{-\beta E_L(x)} := \sum_{\pi(s)=x} e^{-\beta E_C(s)}.$$

Proposition 8.2 (Sparse-energy pushforward). *Assume π is local in the sense that each linguistic coordinate depends on a bounded number of cognitive coordinates, and assume the higher-order cumulants induced by marginalizing over fibers of π are bounded by ε. Then E_L is approximated by a sparse signed pairwise energy*

$$E_L(x) = \frac{1}{2} \sum_{i<j} w_{ij}(1 - s_{ij} x_i x_j) + R(x), \qquad \|R\| \leq O(\varepsilon).$$

Proof sketch. Gibbs pushforward expresses E_L as the negative logarithm of a marginal partition sum. The logarithm admits a cumulant expansion over the linguistic coordinates. Locality of π and sparsity of E_C ensure that low-order terms dominate and that induced interactions have bounded neighborhood size. The assumption on higher-order cumulants bounds the residual R. The pairwise coefficients w_{ij} and signs s_{ij} are the second-order cumulants of the pushed-forward energy. □

Thus narrow word-order universals are the externalization-side pushforward of sparse controller coherence.

8.4 Correspondence 4: graded stability as rate transport

Conceptual version

Universals are not simply true or false. Some are hard-kernel stable; some are middle-shell tendencies; some are outer-shell residue; some are spurious. The linguistic stability coefficient proposed in [1] should be the image of a cognitive stability coefficient under Λ.

Mathematical formulation

Let a cognitive feature p have two-state transition rates

$$0 \xrightarrow{\alpha_C} 1, \qquad 1 \xrightarrow{\beta_C} 0.$$

Let $U = \Lambda(p)$ be the corresponding linguistic feature or universal, with rates α_L, β_L. Define

$$\sigma_C(p) = \log \frac{\alpha_C}{\beta_C}, \qquad \sigma_L(U) = \log \frac{\alpha_L}{\beta_L}.$$

Theorem 8.3 (Stability-rate transport)**.** *Assume Λ distorts transition rates by at most a multiplicative factor $e^{\pm\varepsilon}$:*

$$e^{-\varepsilon} \le \frac{\alpha_L}{\alpha_C} \le e^{\varepsilon}, \qquad e^{-\varepsilon} \le \frac{\beta_L}{\beta_C} \le e^{\varepsilon}.$$

Then

$$|\sigma_L(U) - \sigma_C(p)| \le 2\varepsilon.$$

Proof. Write $\alpha_L = \alpha_C e^{r_\alpha}$ and $\beta_L = \beta_C e^{r_\beta}$, with $|r_\alpha|, |r_\beta| \le \varepsilon$. Then

$$\sigma_L(U) - \sigma_C(p) = \log \frac{\alpha_C e^{r_\alpha}}{\beta_C e^{r_\beta}} - \log \frac{\alpha_C}{\beta_C} = r_\alpha - r_\beta,$$

so the absolute value is at most 2ε. □

This theorem links linguistic stability to cognitive stability. A hard-kernel cognitive invariant, for which β_C is extremely small, maps to a high-stability linguistic universal unless Λ dramatically distorts rates.

8.5 Correspondence 5: context-indexed laws as gluing

Conceptual version

Many universals are local: they hold in subordinate clauses, or in definite object contexts, or in animate-object contexts, or inside a particular phrase type. The cognitive analogue is controller selection: different contexts activate different module supports. The mathematical structure is an indexed family of local closure systems glued over context overlaps.

Mathematical formulation

Let Ctx be a category of contexts. A context-indexed grammar/cognition system is a functor

$$\mathcal{F} : \mathsf{Ctx}^{op} \to \mathsf{Poset}$$

assigning to each context c a feature poset P_c, and to each context refinement $r : d \to c$ a restriction map $r^* : P_c \to P_d$. Suppose each context has a local closure operator

$$\mathrm{cl}_c : \mathcal{P}(P_c) \to \mathcal{P}(P_c).$$

Theorem 8.4 (Exact gluing of local closure laws)**.** *Assume that for every overlap/refinement $r : d \to c$, local closure commutes with restriction:*

$$r^* \, \mathrm{cl}_c(S) = \mathrm{cl}_d(r^* S).$$

Then the family $\{\mathrm{cl}_c\}$ glues to a global closure operator on compatible sections of $\mathcal{F}$.

Proof. A compatible section is a family $(S_c)_c$ satisfying $r^*S_c = S_d$ for every refinement $r : d \to c$. Define

$$\mathrm{cl}((S_c)_c) := (\mathrm{cl}_c(S_c))_c.$$

The commutation assumption ensures compatibility of the resulting family. Extensivity, monotonicity, and idempotence hold componentwise because each cl_c is a closure operator. □

Proposition 8.5 (Approximate gluing defect)**.** *If the commutation equation fails by defects δ_r, then the global closure defect is bounded by the join or sum of the overlap defects, depending on the chosen quantale:*

$$\delta_{\mathrm{glue}} \leq \bigvee_{r \in \mathrm{Mor}(\mathsf{Ctx})} \delta_r.$$

When the defects are induced by cognitive context-update commutators, this is bounded by the corresponding confusion-pair costs from the approximate CCL theorem.

This theorem formalizes why broad aggregation can fail. A local law may be perfectly valid in every context, yet disappear when the context index is forgotten. Forgetting context is a functor that may fail to preserve closure.

9 Semantic frames and GTGI-style cognitive mappings

9.1 Why the frame layer is essential

The functor $\Lambda : \mathsf{Cog}_Q \to \mathsf{Lang}_Q$ factors through Frame_Q because cognitive modules do not usually map directly to typological features. A causal module may support a cluster of person, agency, attention, event, or action schemas. A linguistic feature externalizes only part of that schema, often through an interface that mixes semantics, phonology, and morphology.

The patternist perspective in [9] treats mind as a system of patterns, including patterns in perception, action, memory, inference, attention, and self-modeling. In this paper, semantic frames are the category-theoretic carriers of those patterns insofar as they are relevant to language. They are the objects that make mappings from cognition to language compositional.

9.2 Frame morphisms

A frame morphism

$$r : F \to F'$$

can represent:

- abstraction: a detailed event frame maps to a coarser event frame;
- refinement: a general argument-role frame maps to a more specific case frame;
- embedding: a person frame embeds into a speech-act frame;
- metaphor or transfer: a spatial frame maps to an abstract relational frame;
- externalization preparation: a semantic frame maps to a frame enriched with morphosyntactic slots.

The grade $\omega(r)$ records how much frame structure is preserved and how much cost or ambiguity is introduced.

9.3 Natural transformations among cognitive modalities

The GTGI-style view emphasizes that language is not isolated. It interacts with perception, action, memory, inference, and attention. Categorically, these can be represented as functors into the frame category:

$$\begin{tikzcd}
\mathsf{Perception} \arrow[dr, "P"'] & & \mathsf{Action} \arrow[dl, "A"] \\
& \mathsf{Frame}_Q & \\
\mathsf{Memory} \arrow[ur, "M"] & & \mathsf{Lang}_Q \arrow[ul, "\mathcal{S}"']
\end{tikzcd}$$

Natural transformations between these functors express cognitive synergies: ways in which perception-guided frames, action-guided frames, memory-guided frames, and language-guided frames systematically correspond. The language-to-cognition direction is $\mathcal{S} : \mathsf{Lang}_Q \to \mathsf{Frame}_Q$; the cognition-to-language direction is $\mathcal{E} : \mathsf{Frame}_Q \to \mathsf{Lang}_Q$. Their adjunction formalizes the production/comprehension pairing.

9.4 Sapir–Whorf and its dual

The Sapir–Whorf question asks how language shapes cognition. In this framework that is a question about the composite

$$\mathsf{Lang}_Q \xrightarrow{\mathcal{S}} \mathsf{Frame}_Q \xleftarrow{\mathcal{K}} \mathsf{Cog}_Q.$$

The dual question, central here, asks how cognition shapes language:

$$\mathsf{Cog}_Q \xrightarrow{\mathcal{K}} \mathsf{Frame}_Q \xrightarrow{\mathcal{E}} \mathsf{Lang}_Q.$$

The typological universals record is evidence about the second direction. If a linguistic universal is stable across languages after genealogical and areal correction, then its pullback along Λ is evidence for a stable cognitive frame or invariant. The weakness principle says we should infer the weakest such invariant.

9.5 Three-tier classification of semantic frames

The stability coefficient σ suggests a three-tier classification:

Kernel frames. Frames whose linguistic images have very high stability. Candidate examples include basic person distinctions, basic number distinctions, argument-role accessibility, and elementary event/agent/patient structures.

Consolidated shell frames. Frames whose linguistic images are robust but exceptionful. Candidate examples include many narrow word-order coherence tendencies and moderately stable externalization preferences.

Outer-shell residue. Frames or feature couplings whose apparent universality disappears under correction or holds only in narrow contexts. These are culturally, historically, or locally learned regularities rather than protected substrate.

This classification is not merely linguistic. It is a hypothesis about cognitive architecture, inferred through linguistic direct images and pullbacks.

10 Statistical and computational research programme

10.1 Recover the linguistic subobjects

The first empirical task is to represent each candidate universal as a subobject $U \hookrightarrow L$ in the typological TUG core. This requires estimating:

- the feature lattice or poset in which the universal lives;
- whether the universal is closure-like, energy-like, mediated, stability-like, or context-indexed;
- its empirical stability coefficient $\sigma(U)$, corrected for genealogy and geography;
- its local context restrictions.

10.2 Infer cognitive pullbacks

For each stable linguistic subobject U, infer candidate cognitive pullbacks $\Lambda^*(U)$. Since Λ is not directly observed, this is an inverse problem. The weakness criterion selects the cognitive explanation $E \in \mathrm{Exp}(U)$ maximizing $W(E)$.

Empirically, this means comparing models with different cognitive mediators, different frame structures, and different modular decompositions, and penalizing double-counted dependencies. A broad universal should not be accepted as cognitively real merely because multiple correlated features point in the same direction; it should be accepted when a weak mediator explains the dependency better than redundant correlations.

10.3 Estimate the confusion graph

The approximate CCL theorem bounds forgetting and path-dependence by a graph of confusing context pairs. The linguistic analogue is a graph of contexts or feature domains whose interactions produce violations of factorization. The research task is to estimate this graph from typological data:

$$G_{\mathrm{conf}} = (\mathsf{Ctx}, E_{\mathrm{conf}}, \delta).$$

Edges should correspond to mediators or context overlaps. Sparse confusion is predicted by causal coding; dense confusion would undermine the modularity hypothesis.

10.4 Test closure transport

For hierarchical universals, estimate the cognitive-frame closure relation R_C and the linguistic closure relation R_L. The categorical prediction is that

$$\mathrm{cl}_L \approx \Sigma_\Lambda \, \mathrm{cl}_C \, \Lambda^*.$$

This can be tested by checking whether linguistic closure defects correspond to plausible cognitive-frame defects and whether the closure direction matches non-linguistic cognitive accessibility data.

10.5 Test sparse-energy pushforward

For word-order universals, infer a signed interaction graph over linguistic choices and ask whether it can be represented as a sparse pushforward of a controller-side routing graph. The key predictions are:

- the linguistic graph should be sparse;
- strong hubs should correspond to high-traffic externalization choices;
- diachronic transitions should favor lower energy;
- higher-order residuals should be small but not zero.

10.6 Test stability-rate transport

Estimate $\sigma_L(U)$ for linguistic universals and compare with candidate cognitive stability measures. In neural or cognitive models, analogues include:

- resistance to fine-tuning perturbation;
- low commutator defects across contexts;
- low demotion rates from kernel or inner-shell compartments;
- high reuse utility across tasks.

The prediction is that linguistic high-stability universals correspond to high cognitive stability under these measures.

10.7 Use multilingual neural models as an intermediate probe

Large multilingual neural models provide an intermediate system between human typology and implemented cognitive architectures. They can be probed for:

- modular supports for typological features;
- context-conditioned activation routing;
- commutator defects under sequential fine-tuning;
- stability gradients across parameters or circuits;
- sparse mediators for broad cross-module regularities.

The question is not whether current transformers are human brains. The question is whether systems trained on multilingual data develop internal structures that mirror the categorical transport patterns predicted here.

11 Implications for Universal Grammar and AGI

11.1 Principles and parameters revisited

The old principles-versus-parameters distinction becomes sharper in this framework.

Principles are high-stability closure laws: direct images of cognitive hard-kernel invariants.

Parameters are externalization choices: linguistic images of controller-side routing and sparse coherence pressures.

Mediated broad universals are residual bridge structures: weakest contexts linking modules that would otherwise factor.

Failed universals are often outer-shell residue or context-indexed laws whose context index has been forgotten.

This preserves the useful insight of the generative tradition that some constraints are deep, while incorporating the empirical fact that most typological regularities are graded, context-sensitive, and historically filtered.

11.2 A constraint on AGI architectures

If the correspondences are correct, then a human-level language-capable AGI should not merely learn language statistics. It should instantiate, at some level, the categorical structure that makes linguistic universals stable:

- a causally factorized cognitive substrate;
- context-local updates with small commutators;
- semantic frames mediating cognition and language;
- sparse externalization routing;
- graded kernel-shell stability;
- residual mediators for genuine cross-module laws;
- context-indexed local closure and gluing.

An architecture lacking these structures may still imitate language for many purposes, but it should struggle with lifelong language learning, robust transfer, and systematic generalization across contexts unless it recreates the same structure indirectly.

11.3 Why the typological record matters for cognitive architecture

The typological record has been shaped by many languages, cultures, diachronic changes, and communicative contexts. Once corrected for genealogy and area, stable patterns in that record are not just facts about grammar; they are aggregated evidence about the cognitive machinery that can generate and maintain grammar under continual learning. This makes linguistic universals a potentially powerful indirect probe of cognitive architecture.

12 Limitations and open questions

12.1 Which correspondences are already theorems?

The factorization correspondence is the strongest result: the TUG trace-factorization theorem and the categorical CCL commutator theorem are instances of one monoidal fact. The other correspondences are conditional theorems under explicit assumptions. The main technical work ahead is to determine whether those assumptions can be derived from a common, deeper model of cognition-language externalization.

12.2 How unique is the functor Λ?

The functor Λ is not directly observable. In many cases the more general profunctor $\mathcal{P}$ is more realistic. Different cognitive substrates may support the same linguistic universal, and the same cognitive substrate may externalize differently across languages. Weakness helps choose among explanations, but it does not make the inverse problem trivial.

12.3 Functional and historical explanations

The framework does not deny functional pressures, processing constraints, diachronic drift, or historical contact. Instead, it asks how such pressures interact with the categorical structure of cognition-language externalization. A functional pressure may be the reason a particular energy term exists; a historical process may explain why a language occupies a particular local minimum. The categorical claim is about the shape of the space and the transport of invariants through that space.

12.4 Substrate neutrality

The categories defined here are substrate-neutral. They can describe biological cortical columns, artificial modular neural systems, symbolic or neural-symbolic systems, or other cognitive architectures. The human-brain interpretation is a hypothesis about one realization of the structure, not a requirement of the formalism.

12.5 Open mathematical questions

1. Can the frame category Frame_Q be constructed explicitly from a typed metagraph or distinction-metagraph representation, rather than postulated abstractly?
2. Can the externalization functor $\mathcal{E}$ be learned from multilingual corpora and typological databases?
3. Can the adjunction $\mathcal{E} \dashv \mathcal{S}$ be validated empirically by production/comprehension asymmetries?
4. Can the closure transport theorem be strengthened without the reflectivity assumptions used in Theorem 8.1?
5. Can the sparse-energy pushforward proposition be derived from a concrete HBCML controller rather than assumed as a bounded-cumulant approximation?
6. Can the residual mediator theorem be turned into a practical statistical test for broad universals?
7. Can quantale weakness be estimated robustly from noisy typological and neural data?

13 Conclusion

The category-theoretic formulation changes the flavor of the paper. The central relation between linguistic universals and causal-coding cognition is no longer merely a suggestive five-part analogy. It is a quantale-enriched correspondence between cognitive causal modules, semantic frames, and linguistic TUG cores.

The main conclusion is:

$$\boxed{\begin{array}{c}\text{linguistic universals are direct images of cognitive invariants}\\ \text{along a weakest, quantale-enriched externalization correspondence}\end{array}}$$

and conversely

$$\boxed{\begin{array}{c}\text{cognitive universals are weakest pullback explanations}\\ \text{of stable linguistic subobjects.}\end{array}}$$

The first statement says that stable cognitive structure becomes typological structure when transported through semantic frames and externalized into language. The second says that a stable linguistic universal is evidence for upstream cognitive structure, but only for the weakest such structure needed to explain the data. The weakness clause matters: it is what prevents the theory from multiplying hidden modules, hidden parameters, and redundant correlations beyond necessity.

The strongest theorem is the monoidal transport theorem. Disjoint cognitive supports yield commuting cognitive updates; the cognition-language functor maps this into swap equivalence in the TUG trace quotient; factored cognitive substrates yield factored linguistic cores; and nontrivial broad universals require mediators. This unifies the TUG factorization theorem with the causal-continual-learning commutator theorem.

The other four correspondences take theorem-shaped forms as well. Hierarchical universals arise by closure transport from cognitive-frame closure. Narrow word-order universals arise by sparse-energy pushforward from controller coherence. Graded universalhood arises by stability-rate transport from kernel-shell cognitive dynamics. Context-indexed local laws arise by gluing local closure systems over a context category, with defects bounded by confusion costs.

Quantale weakness supplies the selection principle binding all of this together. TyLAA weakness, causal-coding weakness, and dependence-frontier weakness are not separate metaphors. They are composable layers of one generalized Occam principle: preserve as much admissible indistinction as possible while fitting the data and avoiding unnecessary cost, confusion, and double-counting. This is why causal explanations beat redundant correlation explanations in the framework: they identify the weakest mediator that makes a dependency real.

The resulting research programme is clear. Represent universals as linguistic subobjects. Infer their weakest cognitive pullbacks. Test whether closure, factorization, sparse energy, stability, and context-gluing structures align across typology, cognitive experiments, neural models, and AGI architectures. If they do, the typological record becomes more than a catalog of language facts. It becomes an indirect map of the cognitive architecture that makes human language possible.

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